

Project Initialization and Planning Phase

Date	5 July 2026
Team ID	
Project Title	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution aims to improve agricultural productivity using Machine Learning by recommending the most suitable crop based on soil nutrients and environmental conditions. The system helps farmers make informed decisions, increase crop yield, reduce financial losses, and promote sustainable farming practices through accurate crop prediction..

Project Overview	
Objective	The primary objective is to develop an AI-powered crop recommendation system that predicts the most suitable crop using soil nutrients (N, P, K), pH, temperature, humidity, and rainfall data.
Scope	The project provides farmers with an easy-to-use web application that analyzes agricultural data using machine learning and recommends the best crop for cultivation, improving productivity and supporting sustainable farming.
Problem Statement	
Description	Farmers often face difficulties in selecting the right crop because of limited knowledge of soil nutrients, weather conditions, and crop suitability. This leads to poor crop yield and financial losses.
Impact	Providing accurate crop recommendations will improve crop productivity, reduce farming risks, increase farmers' income, and encourage sustainable agricultural practices.
Proposed Solution	
Approach	Use Machine Learning algorithms to analyze soil nutrients (N, P, K), pH, temperature, humidity, and rainfall, then recommend the most suitable crop through a Flask-based web application..
Key Features	• AI-based crop recommendation system. • Real-time crop prediction using a trained Machine Learning model. • User-friendly web interface for farmers. • Prediction history and recommendation reports. • Responsive design for desktop and mobile devices.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i5 (or higher), 4-Core CPU
Memory	RAM specifications	8 GB RAM
Storage	Disk space for data, models, and logs	256 GB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	Scikit-learn, Pandas, NumPy, Matplotlib, Joblib
Development Environment	IDE	Visual Studio Code, Jupyter Notebook, PyCharm
Data		
Data	Source, size, format	Kaggle Crop Recommendation Dataset, CSV format (approximately 2,200 records with soil nutrients, temperature, humidity, pH, rainfall, and crop labels)